

S3.4 Team Present

A few Recent HCLS AI Data

2026 Spring

[LLM Agents Foundation & Applications](#)

BY Matt and Youke

20260210

Roadmap (three papers)

- 1. Evaluating large language models and agents in healthcare: key challenges in clinical applications
- **2. Holistic evaluation of large language models for medical tasks with MedHELM**
- 3. Tahoe-100M: **A Giga-Scale Single-Cell Perturbation Atlas for Context-Dependent Gene Function and Cellular Modeling**

Clinical Note
Generation

Patient
Communication

Research
Assistance

Clinical
Decision
Support

Admin &
Workflow

1. Paper

Holistic evaluation of large language models for medical tasks with MedHELM

Nature Medicine, 2026

Why Do We Care About Medical LLMs?

Large language models are rapidly being integrated into many domains, but healthcare is fundamentally different from general AI applications.

Medical decisions directly affect patient safety, treatment outcomes, and sometimes survival.

Recent frontier models like GPT-4 have achieved over 90% accuracy on medical licensing exams such as the USMLE.

However, performing well on an exam does not necessarily mean a model can safely operate in real clinical workflows.”

Why do perfect exam scores fail to ensure clinical competence?

The Performance Gap

There remains a substantial gap between high scores on knowledge-based exams and readiness for real-world clinical deployment.

Deceptive Metrics

Success on medical licensing exams inadequately reflects the complexity and diversity of actual clinical practice.

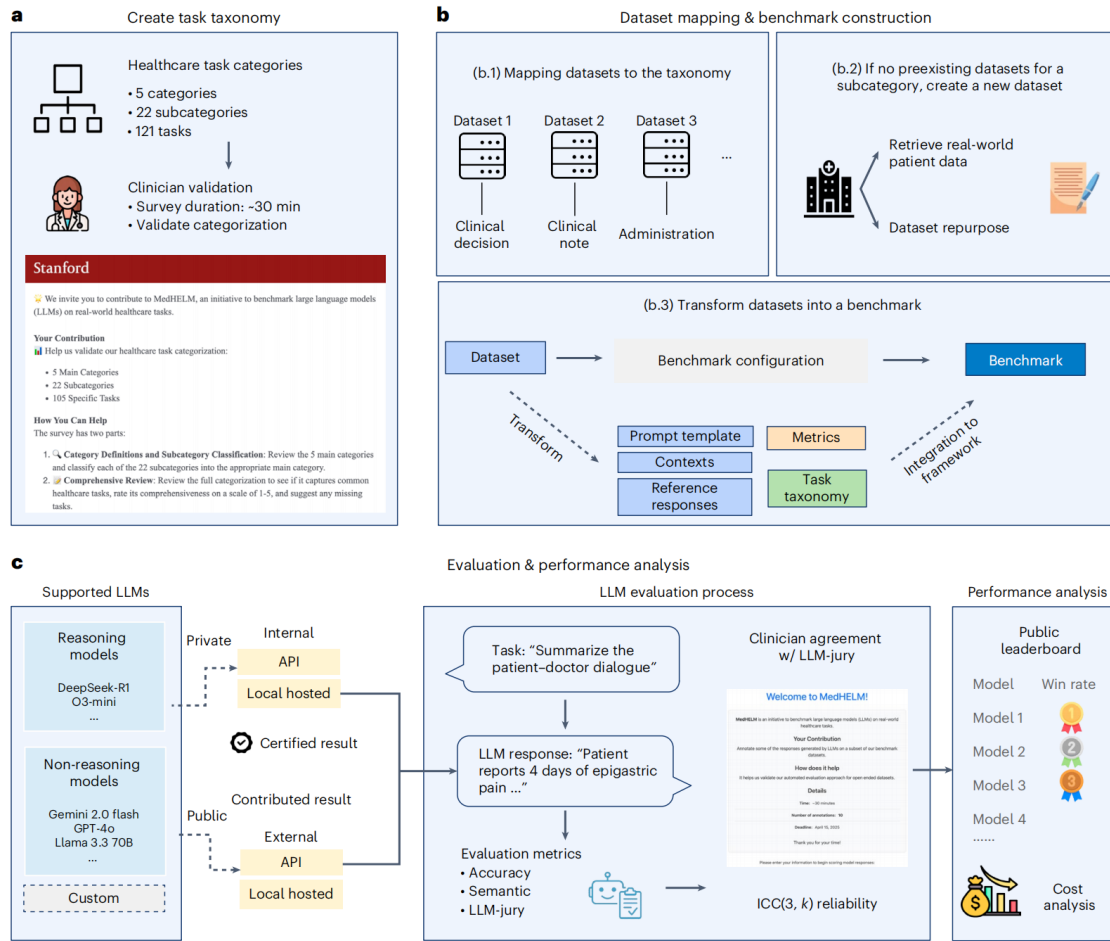
Static vs. Iterative Reality

Current benchmarks primarily use static, multiple-choice questions which fail to capture the iterative, context-aware interactions required by clinicians.

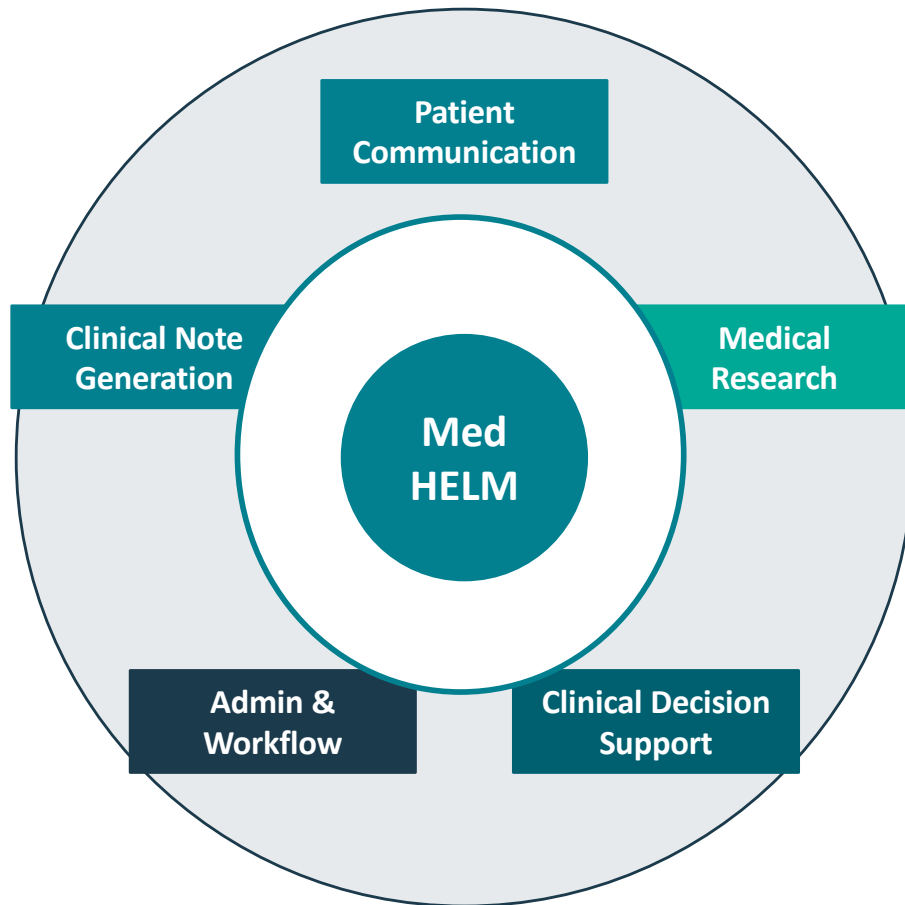
A "Knowledge" Expert, not a "Practice" Copilot

While models excel at retrieving information, they often struggle with the structured tasks that dominate everyday workflows, such as note generation and order reviews.

The MedHELM Pipeline



Mapping the Full Spectrum of Medical Labor



5

Categories

22

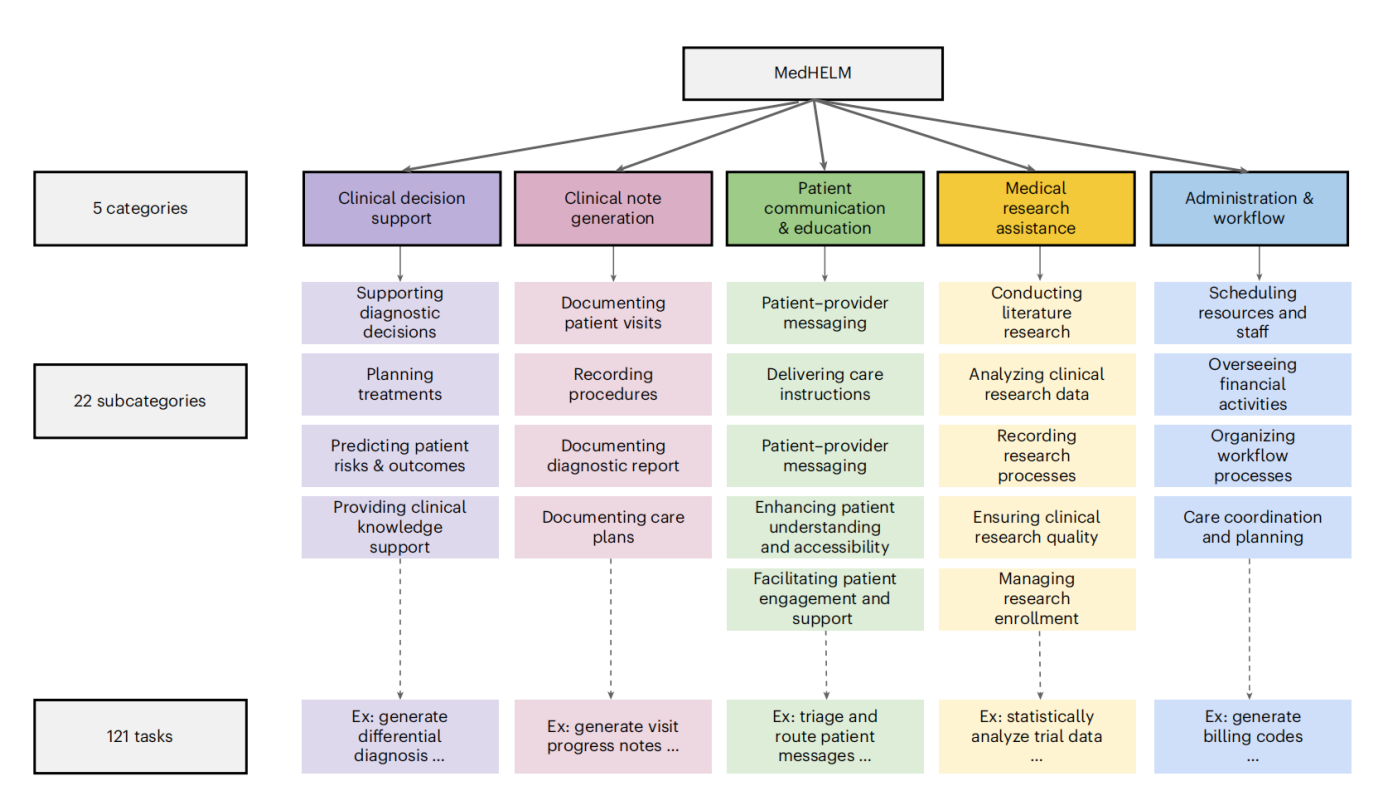
Subcategories

121

Specific Tasks

Structure built and validated by 29 practicing clinicians.

Step 1: Create a Clinician-Validated Task Taxonomy



Step 2: Dataset Mapping & Benchmark Construction

1. Map existing datasets to taxonomy
2. If missing → create new datasets
3. Transform into structured benchmark

Benchmark configuration includes:

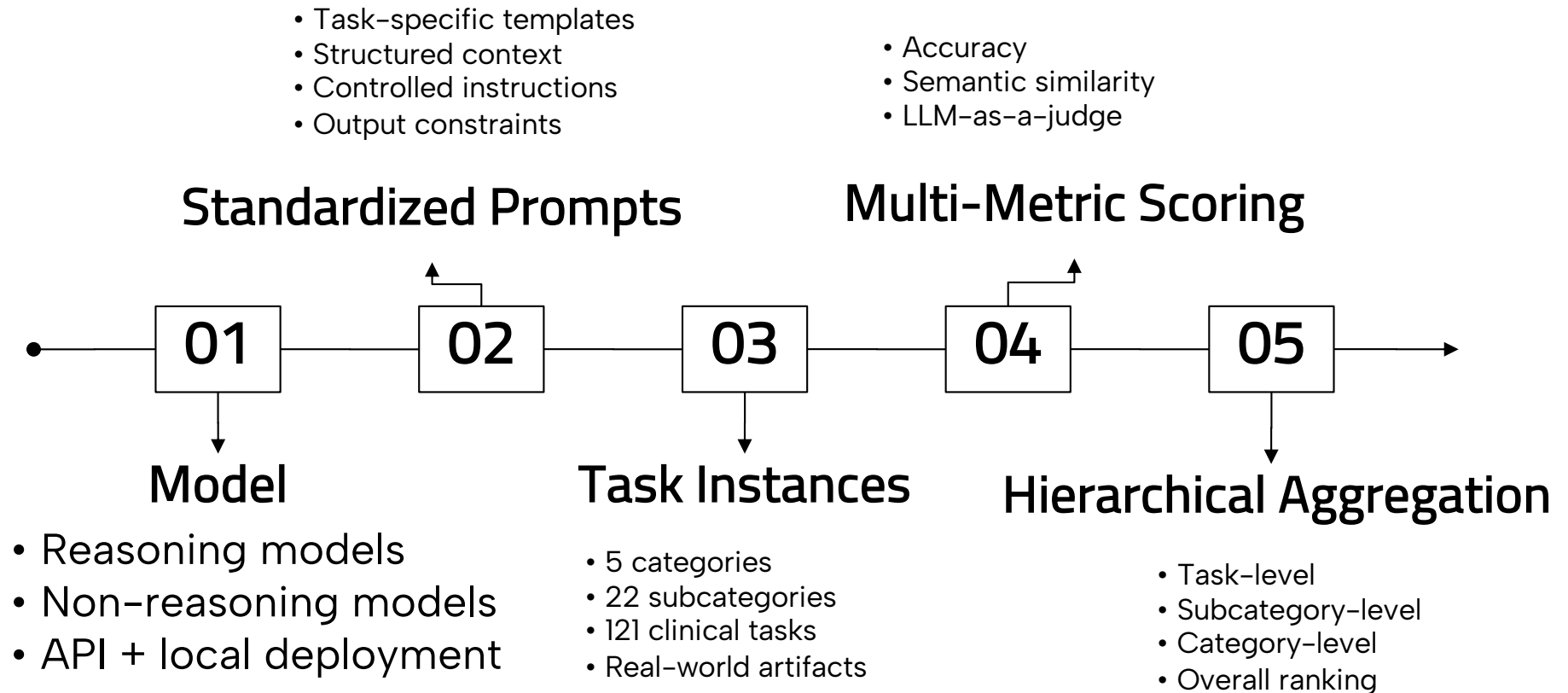
Prompt templates

Context standardization

Reference answers

Evaluation metrics

Step 3: Evaluation Architecture



Example & Workflow

Step1: Task Definition (From Taxonomy)

Category: Clinical Note Generation

Subcategory: Dialogue → Structured Clinical Note

Task Description:

“Summarize the following patient–doctor dialogue into a structured clinical note including history, assessment, and plan.”

Step2: Input (From Benchmark Construction)

Input Context:

Patient–doctor dialogue transcript

Structured output requirement

eg :

Patient: “I’ve had chest pain for two days...”

Doctor: “Does it radiate?”

...

Step4: Evaluation

Multi-layer Evaluation:

Content-level accuracy

Semantic similarity (embedding-based metrics)

LLM-as-a-judge scoring

Clinician agreement validation (ICC(3,k))

Evaluation Dimensions:

Completeness, Clinical correctness

Logical consistency, Professional formatting...

Step3: Model Output

Model Output Includes:

Chief Complaint

History of Present Illness (HPI)

Relevant medical history

Assessment

Plan

Why This Framework Is Different

Traditional Benchmarks:

- Exam-based
- Multiple-choice
- Static datasets
- Knowledge recall

MedHELM:

- Workflow-based
- Task taxonomy-driven
- Real-world alignment
- Multi-metric evaluation
- Reliability validation
- Cost-performance analysis

A Comprehensive Benchmark Suite

To cover the full taxonomy, the team formulated new benchmarks where none existed.

17 Existing Benchmarks

Adapted from prior literature

18 Newly Formulated Benchmarks

Created specifically for MedHELM

35 Total Benchmarks

Covering all 5 Categories defined in the Taxonomy

The 'LLM Jury': Automating Clinical Judgment

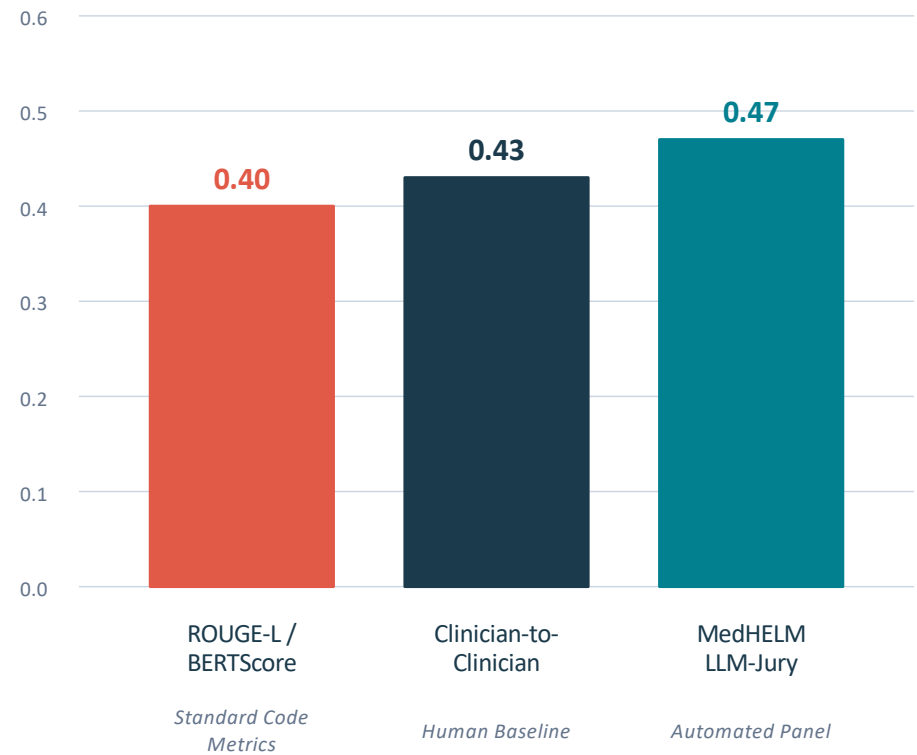
Using a panel of LLMs to grade responses allows for scalable evaluation without losing nuance.

The automated jury is more consistent than a random pair of human doctors.



LLM Jury Panel

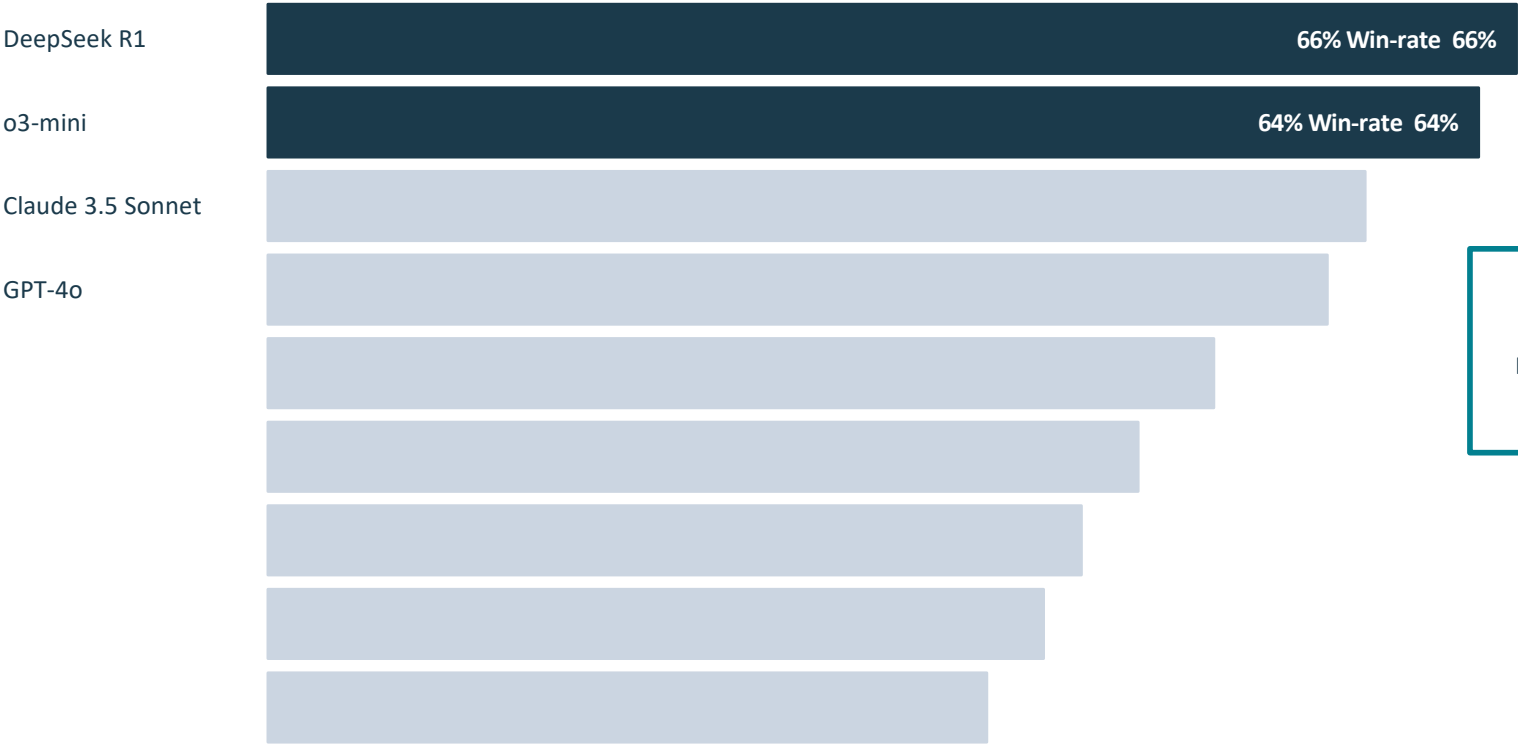
Agreement with Consensus (ICC Score)



Reasoning Models Lead the Pack

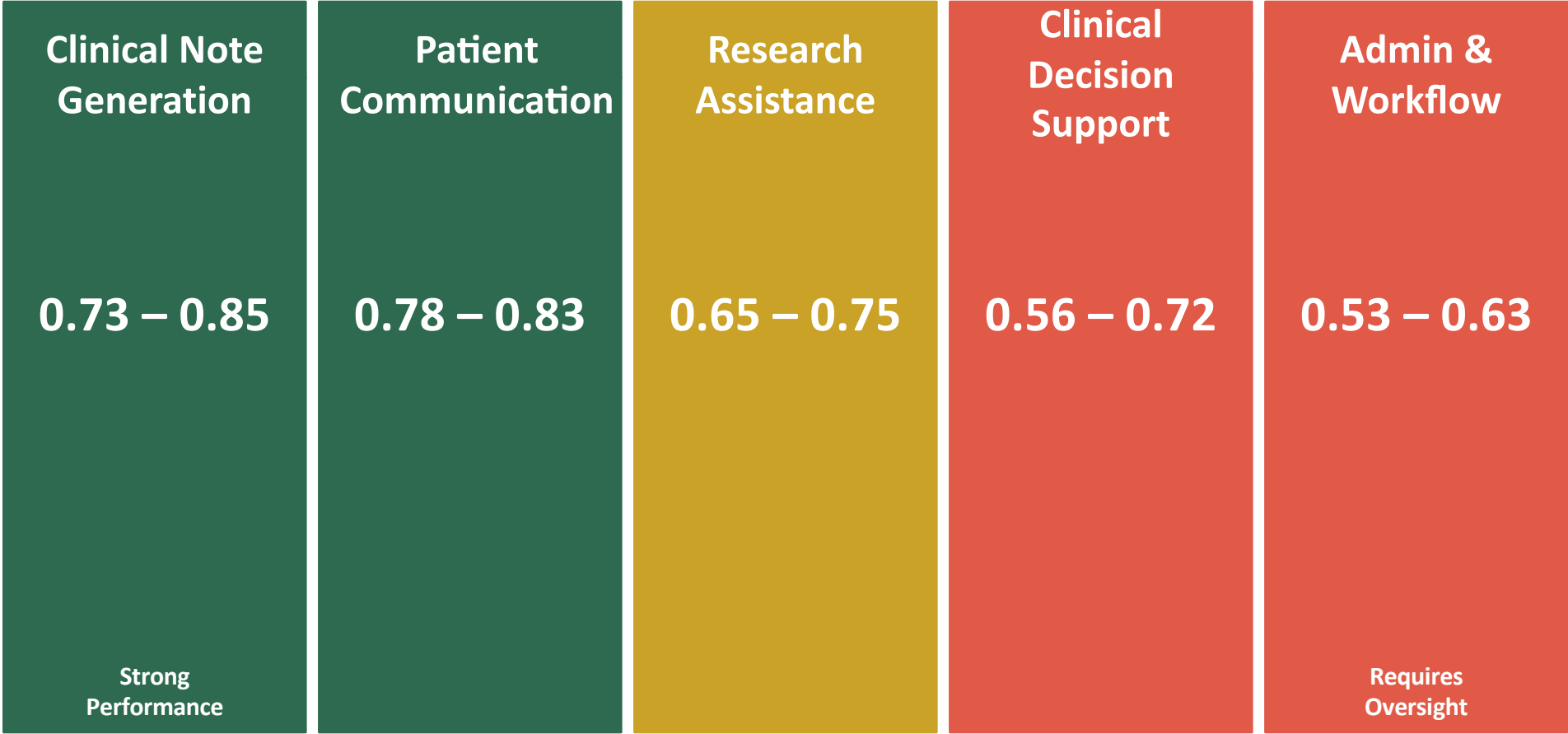
Evaluation of 9 frontier models reveals that success correlates with advanced reasoning capabilities.

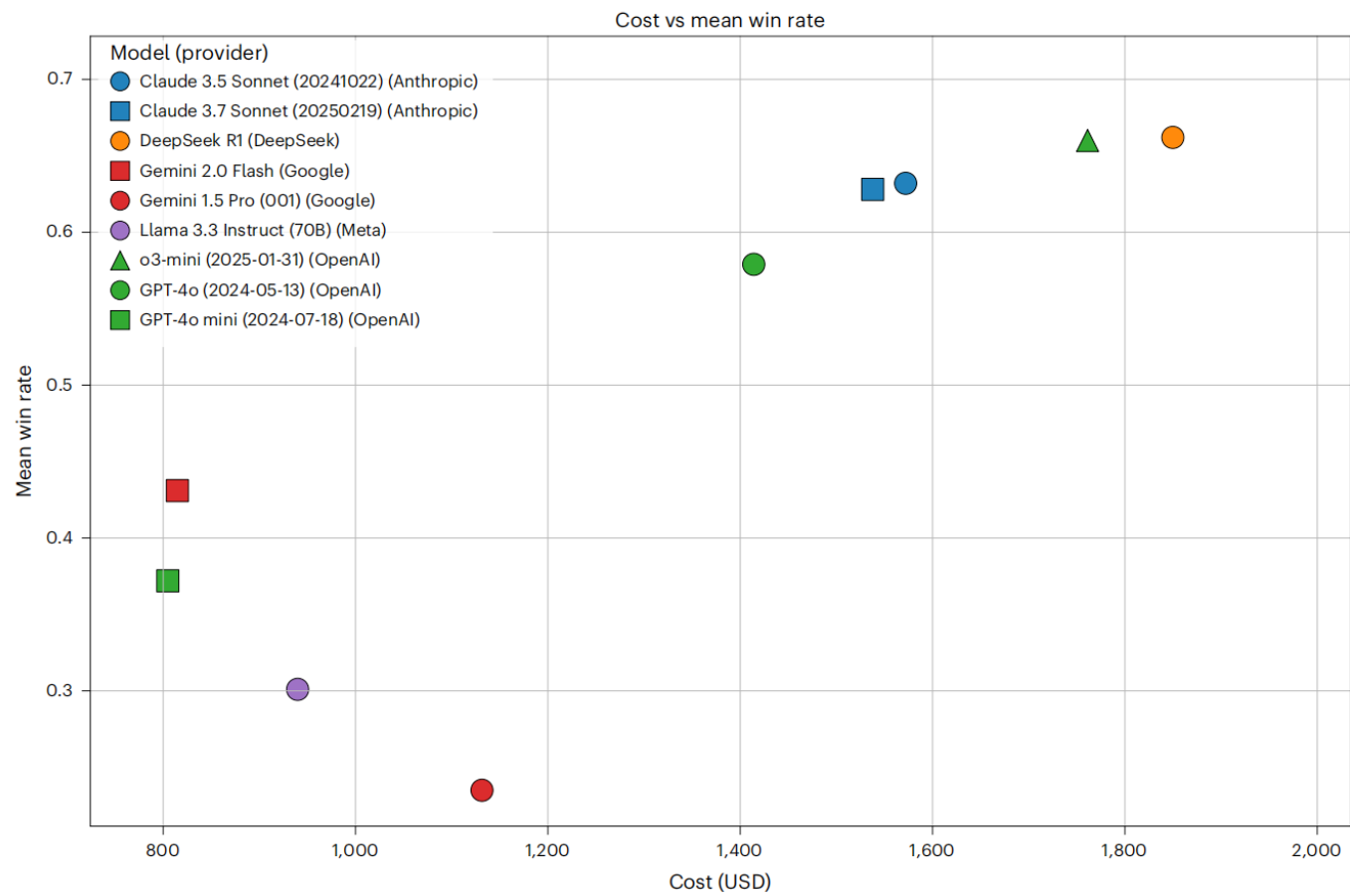
Leaderboard



Models optimized for 'Advanced Reasoning' dominate the leaderboard.

Capability Varies Drastically by **Clinical Category**





Strengths of MedHELM

Moves beyond exam-style benchmarks

Workflow-based evaluation

Clinician-validated taxonomy

Multi-metric scoring

Reliability testing (ICC)

Includes cost-performance analysis

Limitations

Still offline evaluation

No real-time patient interaction

No longitudinal care reasoning

No real accountability

Can We Trust LLM-based Evaluation?

LLM jury reduces human workload

But LLM judges may share biases

Risk of model favoritism

Evaluation drift over time

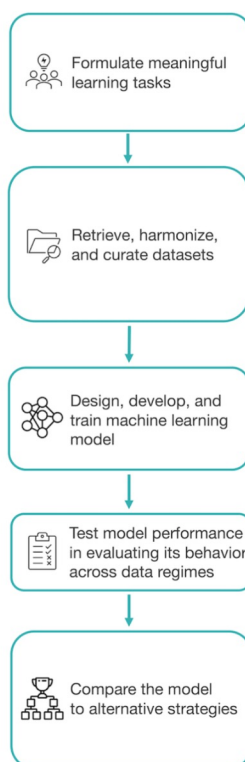
TDC: Therapeutics Data Commons

Presented by Xinyue Xu

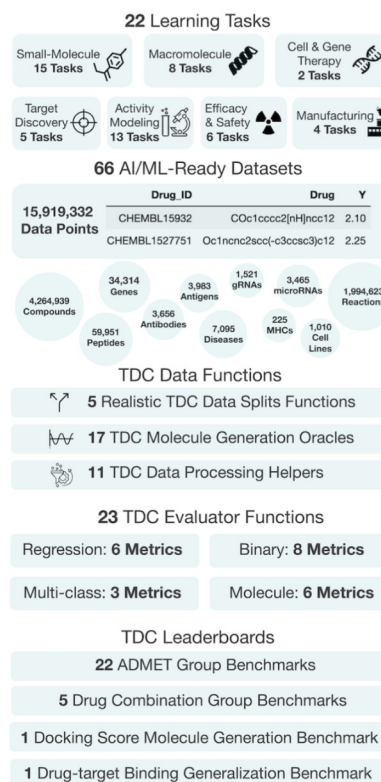
An open science platform that provides:

- Well-defined ML tasks
- AI-ready biomedical datasets
- Data function
- Standardized evaluation protocols
- Public leaderboards

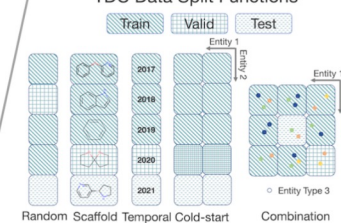
Lifecycle of Therapeutics Machine Learning



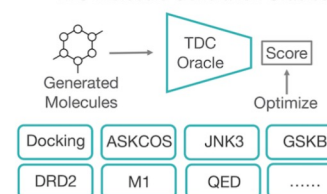
THERAPEUTICS DATA COMMONS



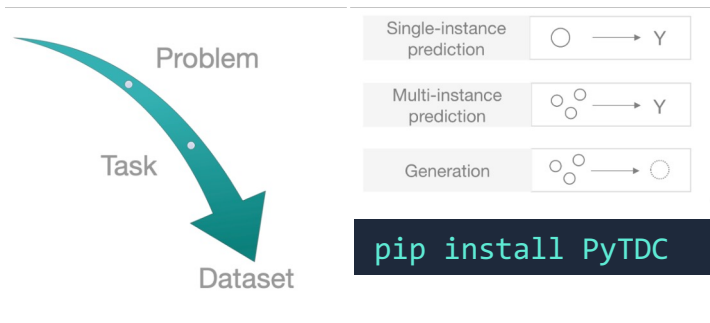
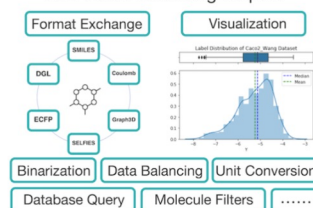
TDC Data Split Functions



TDC Molecule Generation Oracles

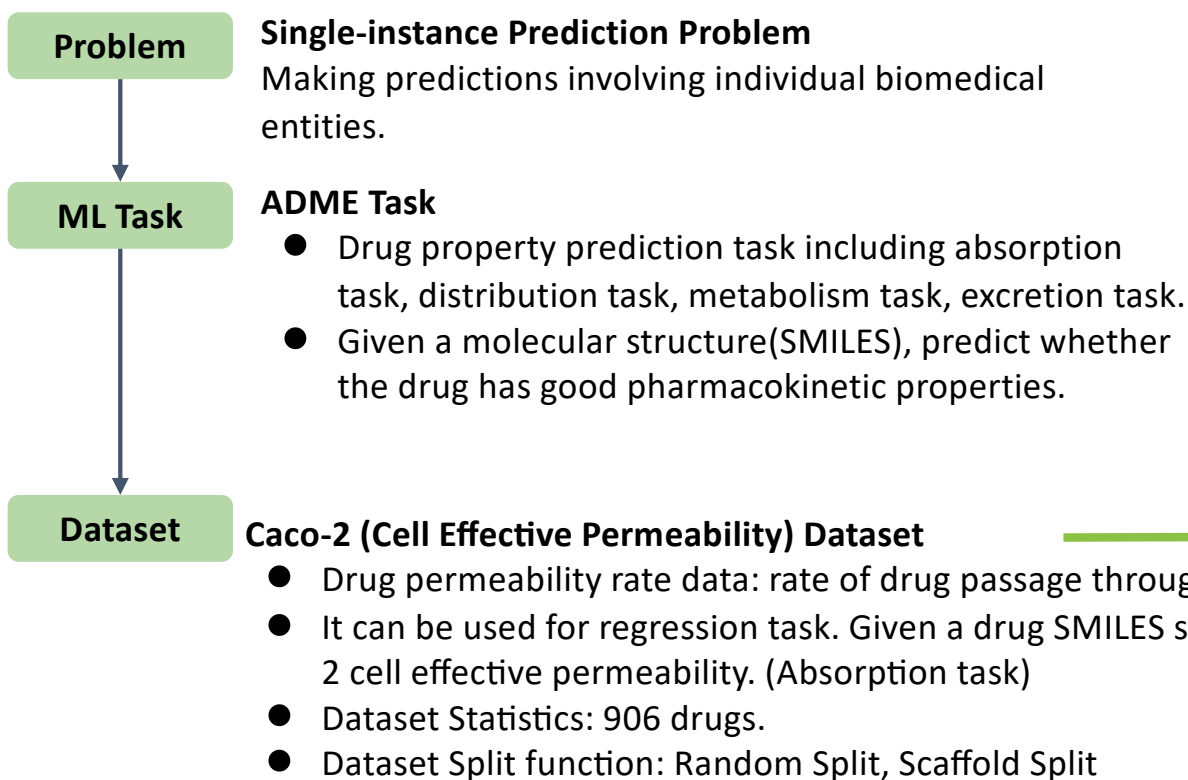


TDC Data Processing Helpers



Example

Presented by Xinyue Xu



Data Loaders Code

```
#install the TDC package  
pip install PyTDC  
#obtain the Caco2 dataset from  
ADME task in the single-instance  
prediction problem  
from tdc.single_pred import ADME  
data = ADME(name = 'Caco2_Wang')  
df = data.get_data()  
splits = data.get_split()
```

TDC vs MedHELM:

TDC

Core Goal:

Benchmark AI for biomedical discovery tasks

Characteristics:

- Focus on molecular / drug discovery tasks
- Structured datasets (graphs, sequences, molecules)
- Predictive modeling (classification / regression)
- Static task definitions
- Emphasis on model generalization

Evaluation Style:

- Accuracy / AUROC / RMSE
- Standard ML metrics
- Dataset-level performance

MedHELM

Core Goal:

Benchmark LLMs for real-world healthcare workflows

Characteristics:

- Focus on clinical task taxonomy
- Workflow-aligned tasks
- Clinical notes & dialogue
- Administrative tasks
- Real-world artifacts

Evaluation Style:

- Accuracy + semantic similarity
- LLM-as-a-judge
- Clinician agreement (ICC)
- Cost-performance analysis

TDC → Measures scientific prediction capability

MedHELM → Measures clinical workflow capability

Dimension	TDC	MedHELM
Domain	Biomedical research	Clinical practice
Task Type	Predictive modeling	Workflow execution
Data	Structured	Semi-structured /unstructured
Interaction	Single-task	Multi-step workflow
Safety context	Indirect	Direct patient impact

Presented by Matt Juntima

Review

Evaluating large language models and agents in healthcare: key challenges in clinical applications

Xiaolan Chen¹ #, Jiayang Xiang² #, Shanfu Lu³ #, Yexin Liu⁴, Mingguang He^{1 5 6}  , Danli Shi¹
²  

2. Evaluating large language models and agents in healthcare: key challenges in clinical applications

Background/Motivation

- LLMs are very useful in conversations and have been applied to many different industries
- Ideas with integrating LLM agents to the healthcare space
- If LLM agents has hallucinations in healthcare, could be dangerous for anyone involved
- Currently no good way of evaluating if certain LLM agents are safe for medical use
- Patients, doctors, administrators, and many more stakeholders will benefit with better evaluated LLM agents.



Problem

1. Data bias - datasets from specific domains or populations
2. Evaluations across diverse healthcare applications lack depth - fails to differentiate strengths and weaknesses of LLMs in practical scenarios
3. Prior evaluation methods primarily focus on accuracy, but overlooks critical things hallucinations, logical reasoning, and the likelihood of generating harmful content



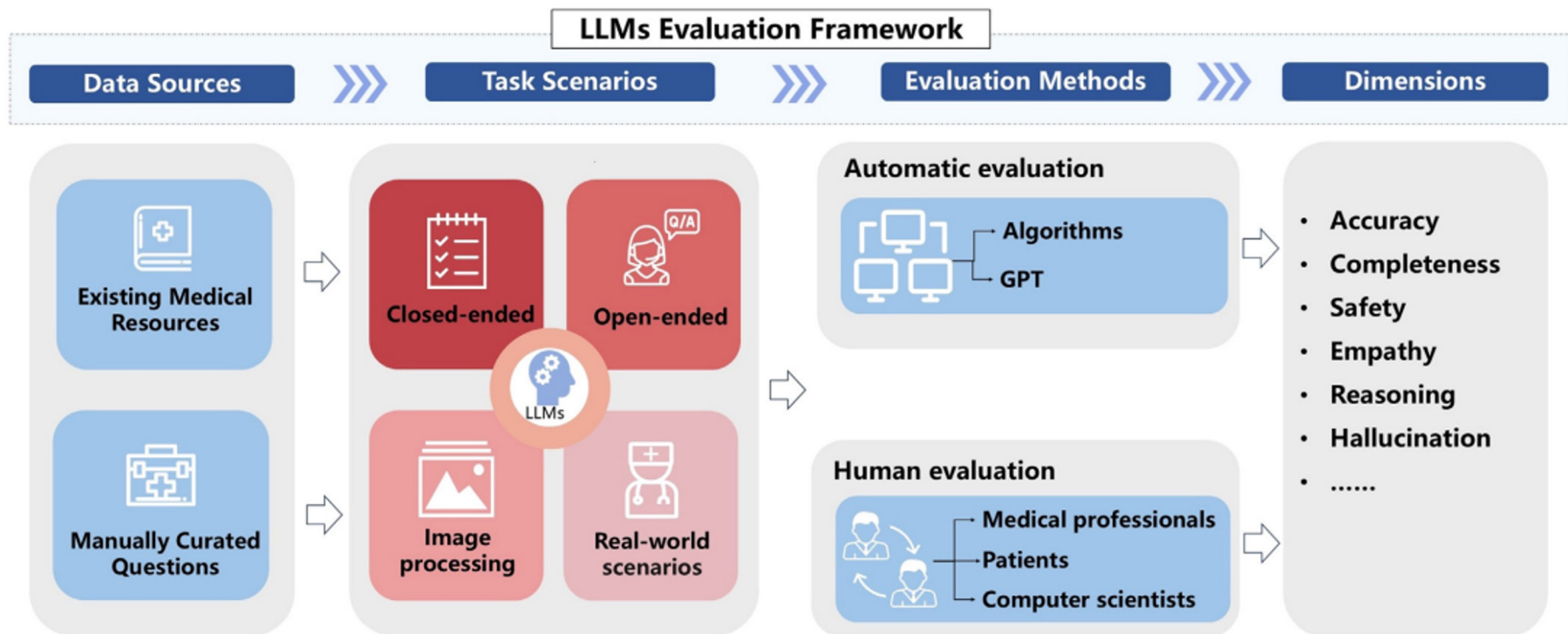
Solution

1. Expanding of data sources and task scenarios
2. Create diverse evaluation methodologies from objective accuracy metrics to more nuanced human-centric evaluations
3. Identified key challenges and opportunities in medical LLM and LLM agent evaluation

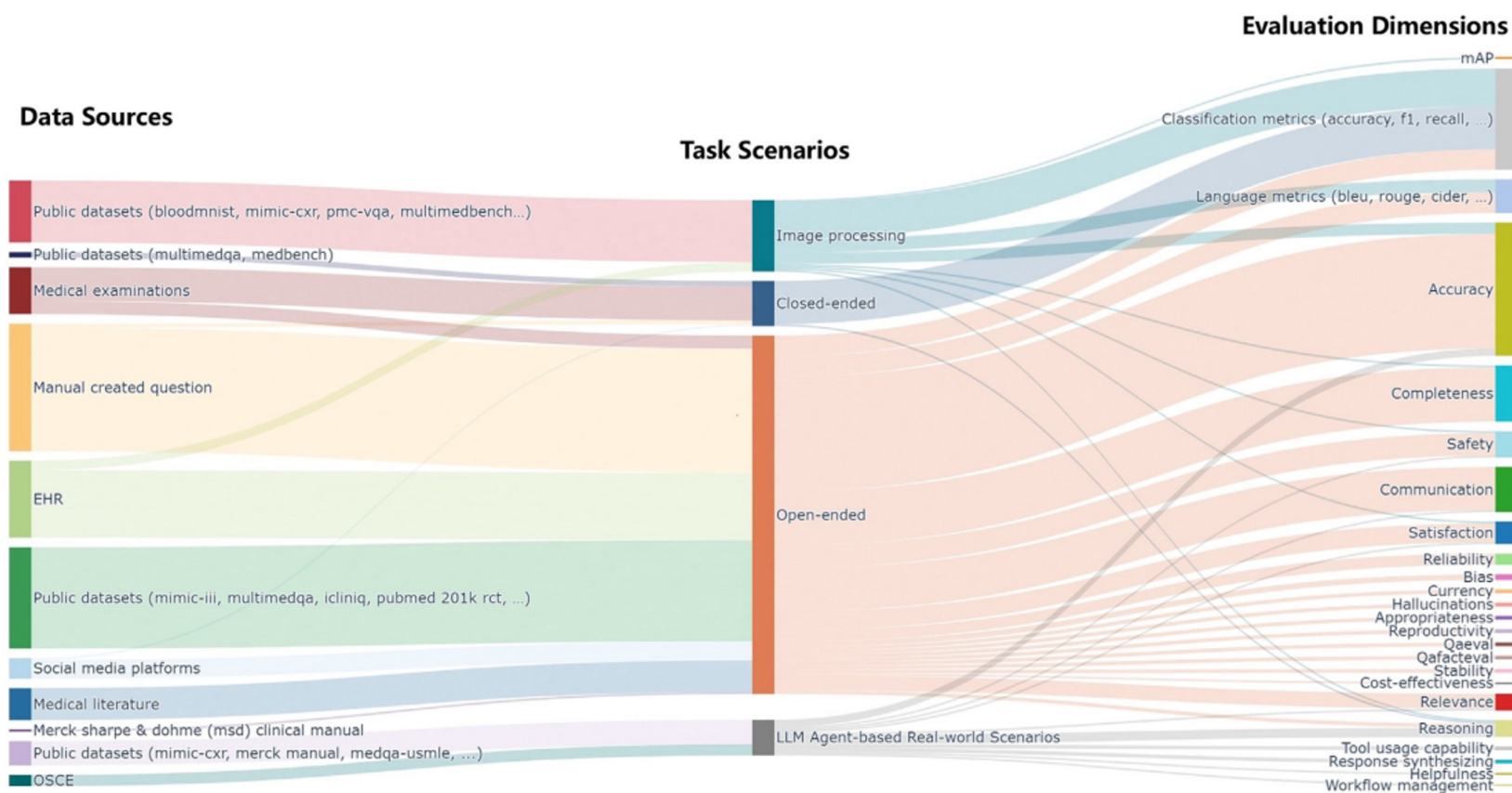
Experimentation Methodology

- Systematic search was carried out across PubMed, Google Scholar, and Web of Science databases between 2023-2024
- Datasets can be broadly divided into 2 main categories: existing medical resources and manually curated question sets
- Different tasks such as close-ended and open-ended tasks, summarizing, information extraction, etc.

Solution



Summary of Data Sources/Evaluation Dimensions



Tasks scenarios

- **Close-ended tasks**
- **Open-ended tasks:** Summarization, Information extraction, Medical question answering
- **Image processing tasks:** Image classification, Report generation, Visual question answering, Other scenarios
- **Real-world multitasking involving LLM agents**



A

Text prompt: Please judge the disease type shown in this chest X-ray:
A. Normal
B. Pneumonia
C. Pulmonary tuberculosis
D. Lung cancer



B

Text prompt: Please judge the whether this dermoscopy image contains melanoma:
A. Yes
B. No
C. Uncertain

Image classification

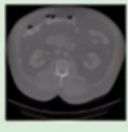


C

Text prompt: Please write a radiology report based on this chest X-ray image, describing the main findings and providing diagnostic opinions.

Reply: The chest X-ray reveals no evidence of acute cardiopulmonary disease. Specifically, the heart, pulmonary vasculature, and mediastinum are within normal limits. Furthermore, there is no indication of pleural effusion or pneumothorax, nor are there any focal areas of consolidation suggestive of pneumonia.

Image report generation



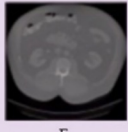
D

Question 1: What imaging modality was used to capture this image?
Reply: CT.

Question 2: Are there any abnormalities visible on the current image?
Reply: Yes, there is a mass-forming lesion within the pancreas...

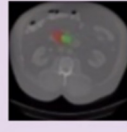
Question 3: I often feel some pain in my lower left abdomen. What could be causing this?
Reply: According to the image, there may be possibility of a pancreatic tumor.

Visual question answering



E

Text prompt: According to the text prompt, segment the corresponding organs and tumors. Pancreas: A nodular organ in the abdomen that contains a mixture of endocrine glands and exocrine glands. Pancreatic tumor typically presents as mass-forming lesions within the pancreas.



Others (image segmentation)

Closed Ended Tasks

- Evaluated models such as GPT-4 on the CMedExam dataset and found that its **61.6%** accuracy was still significantly lower than human level (**71.6%**)
- GPT-3 and GPT-4 evaluated with the American College of Gastroenterology self-assessment tests. A score of **70%** or higher is generally required to pass the evaluation. ChatGPT-3.5 and ChatGPT-4 scored **65.1%** and **62.4%**, respectively.
- Closed ended tasks are easy to test but are not complex

Open Ended Tasks

Divided into **summarization**, **information extraction**, and **medical QA**:

- **Summarization:**

- Automatic metrics were used to assess the lexical and semantic similarity, and human evaluation was conducted on coherence, factual consistency, comprehensiveness, and harmfulness
- LLMs could be susceptible to generating incorrect information and lead to potential harm due to misinformation
- Novel feasibility of using LLMs to generate patient-friendly summaries of radiology reports

Open Ended Tasks Continued

Information Extraction:

- Numerous studies showed LLMs have the potential to assist researchers or patients search and acquire knowledge from a large amount of biomedical data
- ChIP-GPT had a 90%–94% accuracy when trained with 100 examples.

Medical question answering:

- Performance of ChatGPT - considered average
- Models need to be specifically trained for individual medical subspecialties
- ChatGPT demonstrates empathy when answering patients' questions.

Image processing tasks

Divided into **image classification**, **report generation**, **visual QA**, and **other scenarios**

Image classification:

- The results showed that LLMs performed well on different medical image tasks but still need improvement in recognizing fine-grained visual concepts
- Results indicated that while CNN models excelled due to specialized training, vision-language models (VLMs) showed promise in zero-shot and few-shot scenarios without prior training

Image processing tasks

Report Generation:

- Model could generate descriptive reports with structured prompts, but there is still space for improvement in the accuracy of generating specific medical terms

Visual QA:

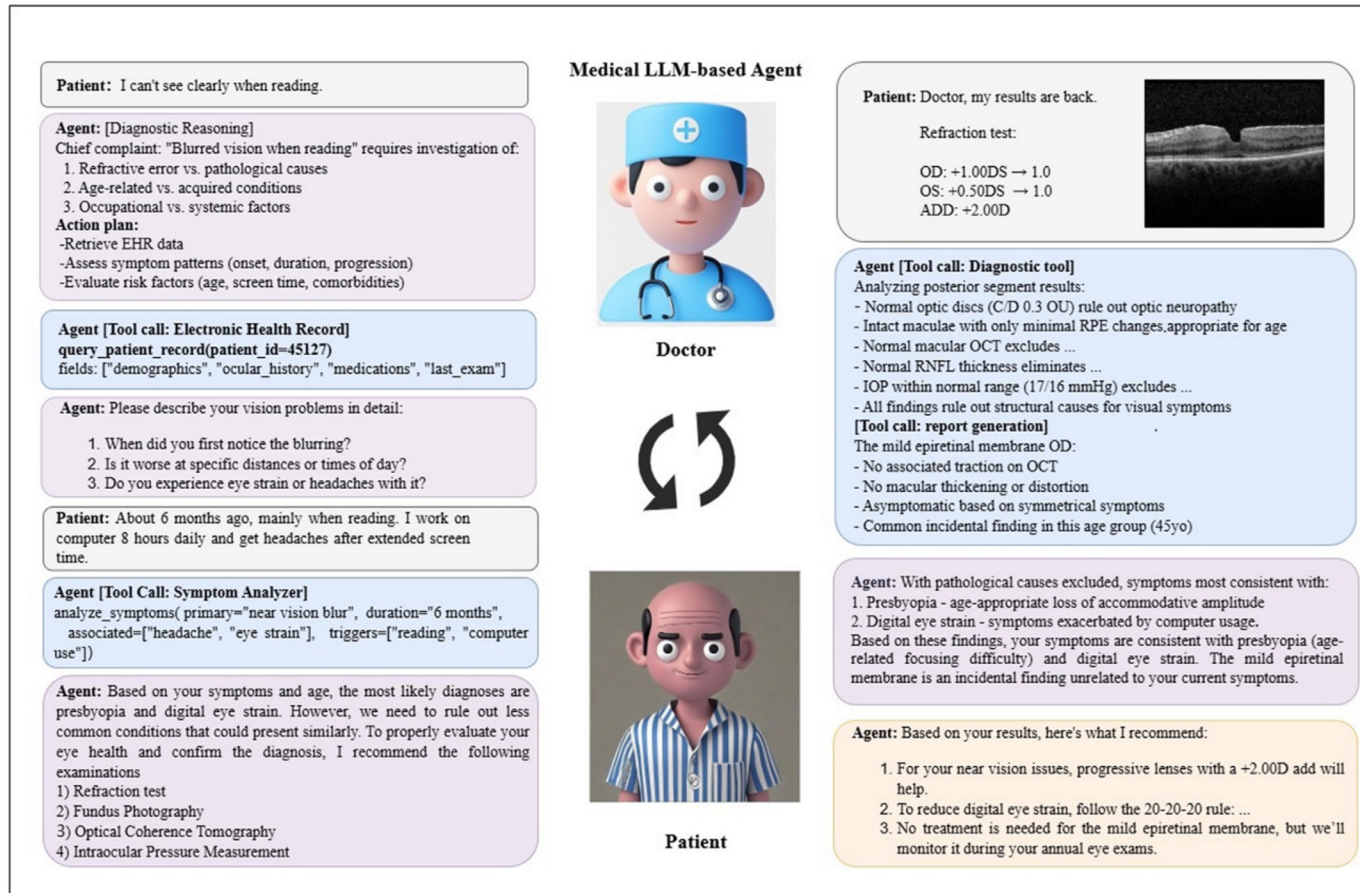
- Requires stronger cross-modal understanding capabilities and higher demands on natural language expression.
- Constructing large scale, high-quality datasets is crucial for evaluating model performance.

Image processing tasks

Other Scenarios:

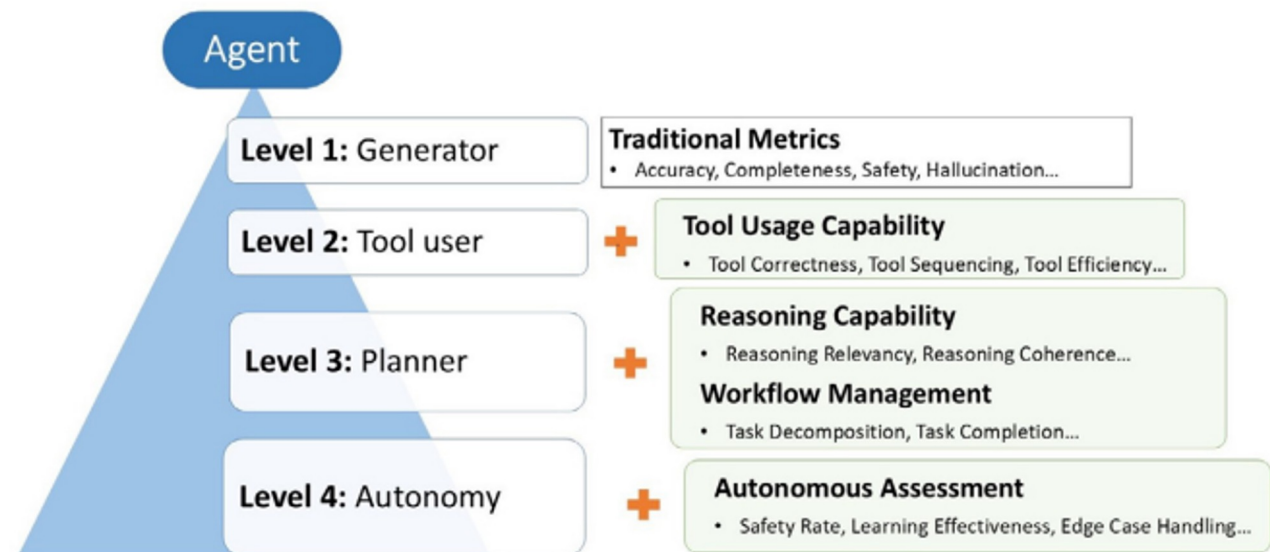
- LLMs also show good prospects in other medical image analysis tasks such as medical image segmentation and cross-modal retrieval
- Current evaluations of LLMs in these new tasks are inadequate

Real-world Multitask Example

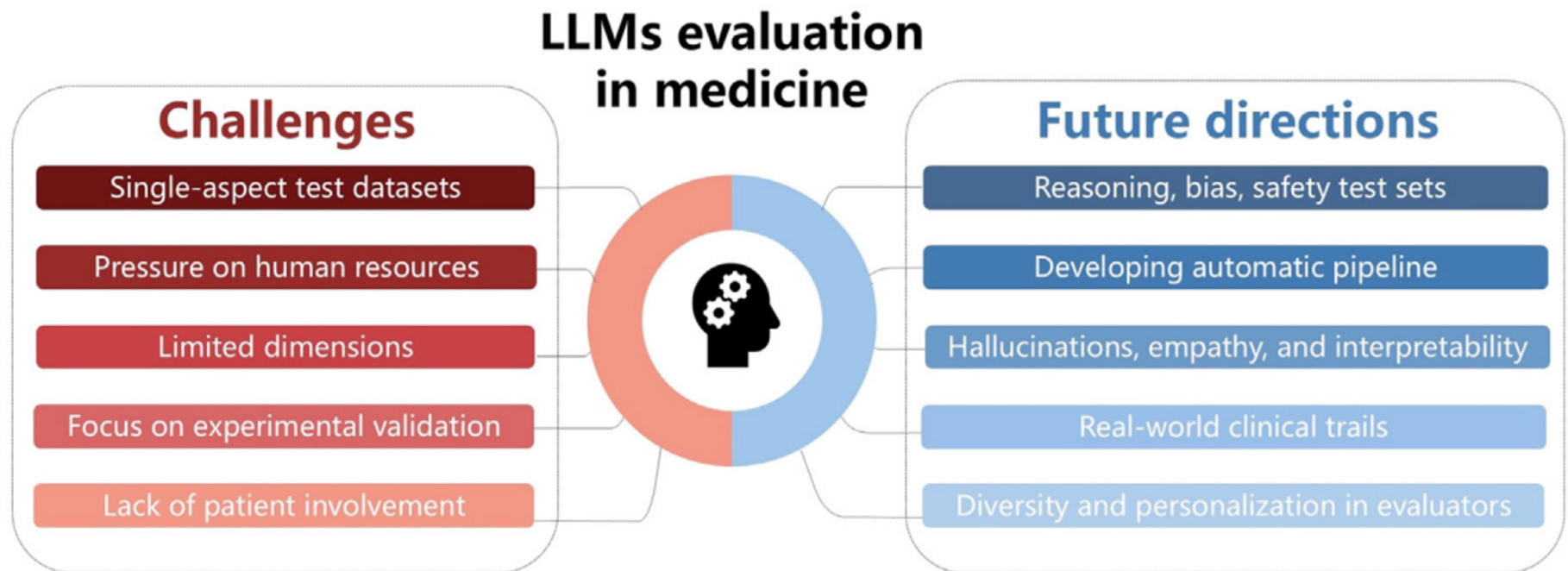


Evaluation Methodology

- Traditional evaluation dimensions - accuracy
- LLM agent evaluation dimensions



Challenges/Future Directions



Contributions

1. Summarized data sources used in evaluations - includes existing medical resources and manually designed clinical questions
2. Analyzed key medical task scenarios
3. Compared evaluation methods and dimensions



Conclusion

- Paper argues for a precise and effective evaluation framework, current standards are not good enough
- The requirements for a good framework is what follows:
 1. Using high quality datasets
 2. Combining human evaluation with automated evaluation
 3. Setting comprehensive and effective assessment standards beyond accuracy
 4. Including specific LLM agent evaluation methods
 5. Having future trial designs being more aligned with real-world clinics and compare LLMs to existing practices

3. Paper: Tahoe-100M

A Giga-Scale Single-Cell Perturbation Atlas for Context-Dependent Gene Function and Cellular Modeling

Zhang et al., 2025 · Vevo Therapeutics & Arc Institute
bioRxiv · doi: 10.1101/2025.02.20.639398

Presentation Outline

① Motivation

Why do we need giga-scale single-cell perturbation data?

② Method Details

The Mosaic platform and experimental design

③ Explanations

Key concepts: cell villages, scRNA-seq, demultiplexing

④ Experimental Setup

14 plates · 50 cell lines · 1,135 drug-dose conditions

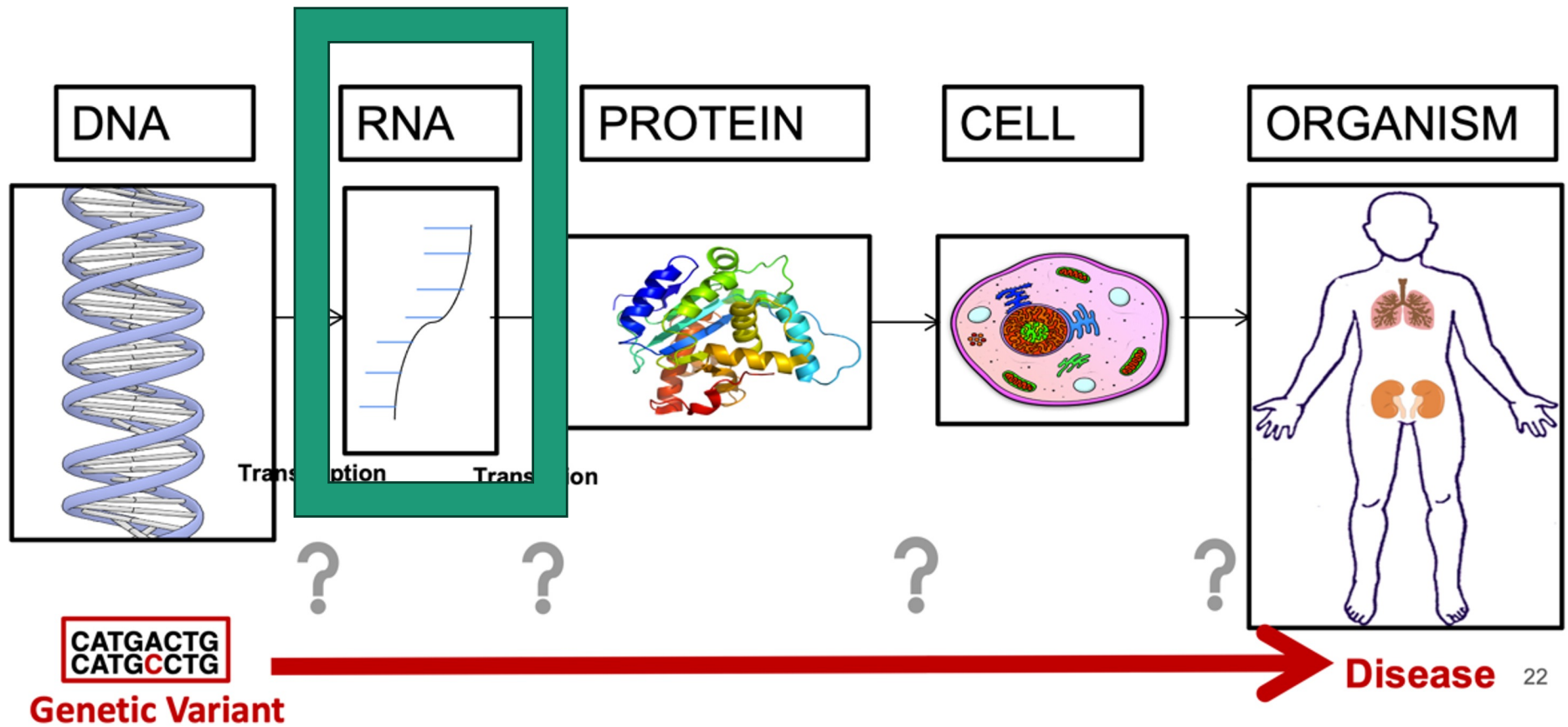
⑤ Results

- Atlas-scale transcriptomics
- Drug effect landscape (E-distance)
- Context-dependent RAS/RAF biology
- Cell cycle modulation

⑥ Takeaways

Implications for AI cell modeling and cancer biology

Biology in a Slide:



The Grand Challenge: Modeling the Cell

A longstanding goal in cell biology:

Build *predictive, mechanistic* models of how cells integrate signals and execute transcriptional responses.

What this requires:

- 1 Large-scale, quantitative datasets of cellular states
- 2 Systematic mapping of how interventions reshape those states
- 3 AI/ML models trained on this data to generalize predictions

Perturbative measurements reveal causal gene-gene interactions, feedback circuits, and compensatory pathways – the regulatory network of the cell.

Conceptual Figure

Diagram: Cell state space $\rightarrow$ perturbation $\rightarrow$ new state
(Illustrate: input signal, transcriptional response, AI model)

Adapted from Bunne et al., 2024 (Cell)

How to Build the Virtual Cell with Artificial Intelligence: Priorities and Opportunities, cell 2024

Why Single-Cell? Why Not Bulk?

Bulk RNA-seq limitations

- Averages across millions of cells
- **Masks critical heterogeneity:** cell cycle phase, transcriptional substates, genetic background
- Cannot distinguish responders from non-responders
- Loses subpopulation dynamics under drug treatment

Single-cell RNA-seq (scRNA-seq) advantages

- Molecular portrait of **each individual cell**
- Resolves heterogeneous drug responses
- Captures rare subpopulations
- Enables cell cycle, trajectory, and clustering analyses

Key insight: Even genetically identical cells respond differently to the same drug. Bulk approaches miss this entirely.

Buettner et al., 2015; Trapnell et al., 2014; Macaulay et al., 2016

What Is Single-Cell RNA Sequencing?

Traditional RNA-seq → bulk average across millions of cells

scRNA-seq → gene expression of each individual cell

Analogy

Bulk RNA-seq is like blending a fruit salad and tasting the mix. scRNA-seq is tasting each fruit separately — you learn what makes each one unique.



Cell Heterogeneity

Reveals cell-to-cell differences hidden in bulk data



Spatial Transcriptomics

Maps gene expression to tissue locations



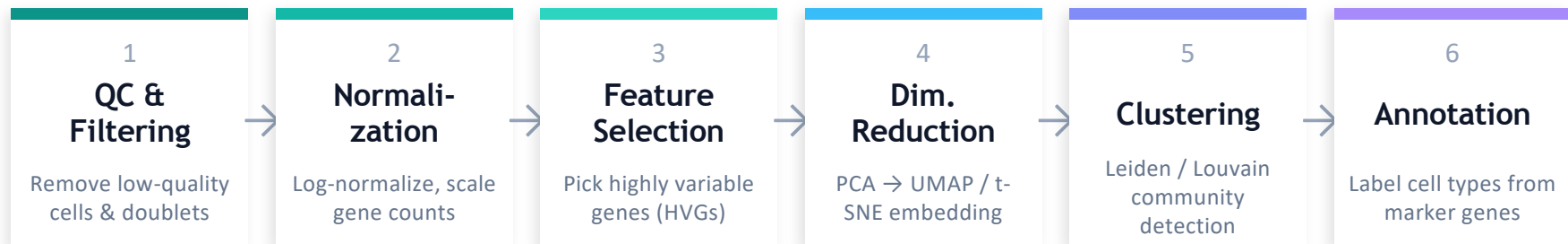
Massive Scale

Thousands to millions of cells per experiment

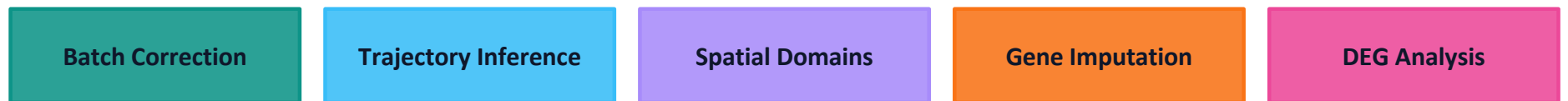
These technologies have produced massive datasets requiring advanced computational pipelines.

The Traditional scRNA-seq Pipeline

Every analysis follows a similar chain of steps — each requiring tool choices and parameter tuning.



Downstream Tasks



⚠ Each step needs tool selection + parameter tuning + biology expertise + Python coding

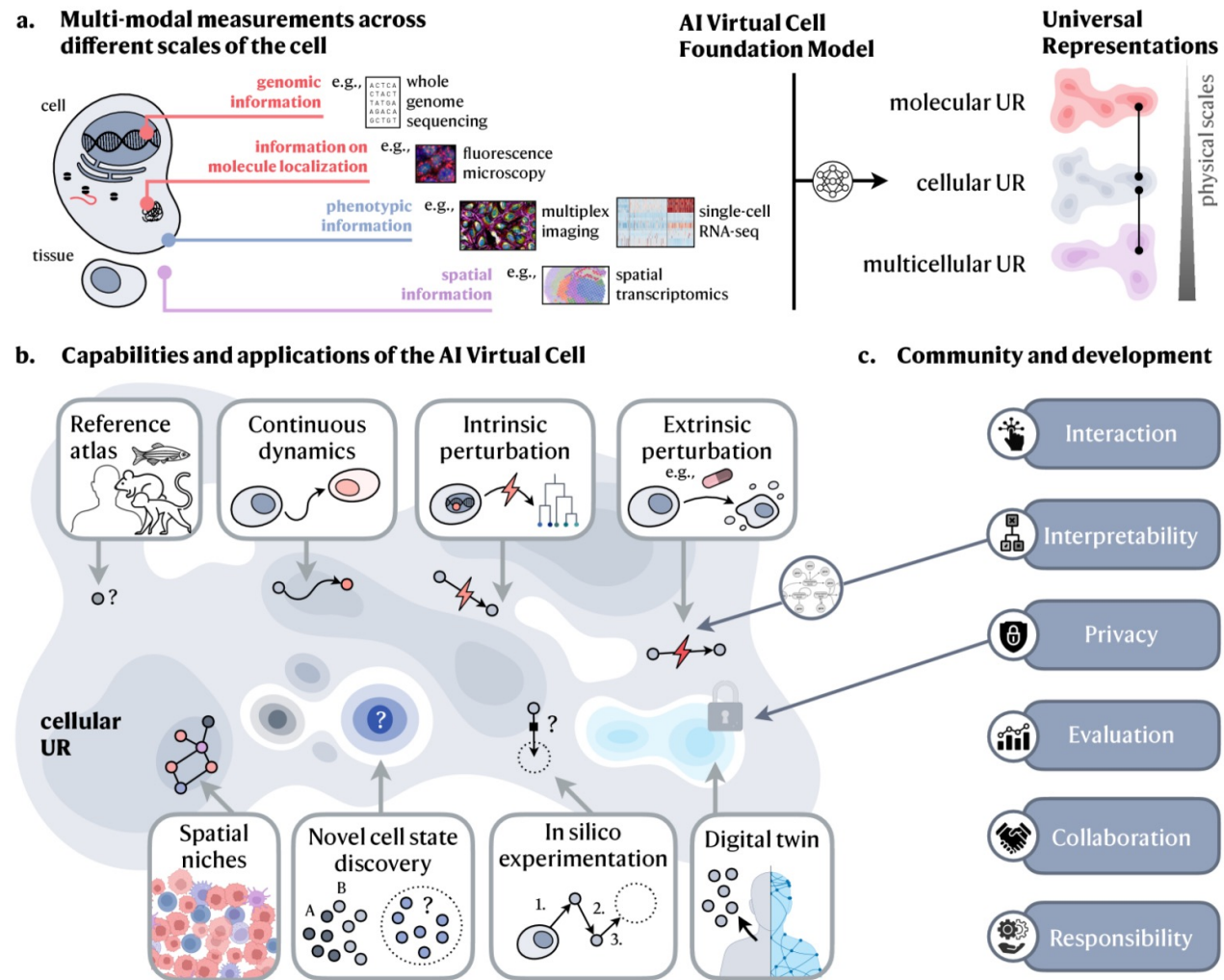


Figure 1: **Capabilities of the AI Virtual Cell.** a. The AI Virtual Cell provides a Universal Representation of a cell state

Why Single-Cell? The Limits of Bulk Approaches

Bulk RNA-seq — Limitations

- Averages across millions of cells
- Masks critical heterogeneity: cell cycle phase, transcriptional substates
- Cannot distinguish responders from non-responders
- Loses subpopulation dynamics under treatment

Single-Cell RNA-seq — Advantages

- Molecular portrait of each individual cell
- Resolves heterogeneous drug responses within a population
- Captures rare resistant subpopulations
- Enables cell cycle, trajectory, and clustering analyses

The Landscape Gap: No Atlas at Scale

Dataset	Cells	Cell Lines	Conditions	Limitation
SciPlex-3 (2020)	~650K	3	~188 drugs	Very few cell lines
Replogle Perturb-seq (2022)	~2.5M	2	~10K genes	Genetic only; 2 lines
Tahoe-100M (2025)	>100M	50	1,135 drug-dose	None — this is the goal

Why existing data falls short:

- Multi-study aggregations suffer from inconsistent protocols and severe batch effects
- Most datasets focus on one cell line — cannot reveal how genetic background shapes drug response
- Coverage too narrow: AI models need diverse training data to learn general regulatory logic

The Mosaic Platform: Cell Villages at Scale

A 'cell village' pools 50 cancer cell lines in one well, so all lines experience identical drug conditions simultaneously.

- 1 Pool 50 cell lines** Equal-ratio Mosaic tumor spheroids in 96-well plates (14 plates total)
- 2 Apply 1,135 drug-dose conditions** 24 h treatment; DMSO vehicle controls in wells H11/H12 on every plate
- 3 Dissociate & fix cells** TrypLE Express + Parse Evercode Cell Fixation v3; plate coordinate identity maintained
- 4 Combinatorial barcoding** Parse GigaLab: 3 rounds of split-pool ligation → ~100K cells per sublibrary; 1,786 sublibraries total
- 5 Sequence** Ultima Genomics UG-100 — 1.4 trillion raw reads; 153 million cells pre-QC
- 6 Bioinformatic deconvolution** split-pipe → demuxlet SNP deconvolution → drug annotation → QC → AnnData

Tahoe-100M: The Answer

Tahoe-100M is the *largest publicly available single-cell dataset to date* – over **100 million transcriptomes** from 50 cancer cell lines treated with 1,100+ small-molecule perturbations, generated on a single unified platform.

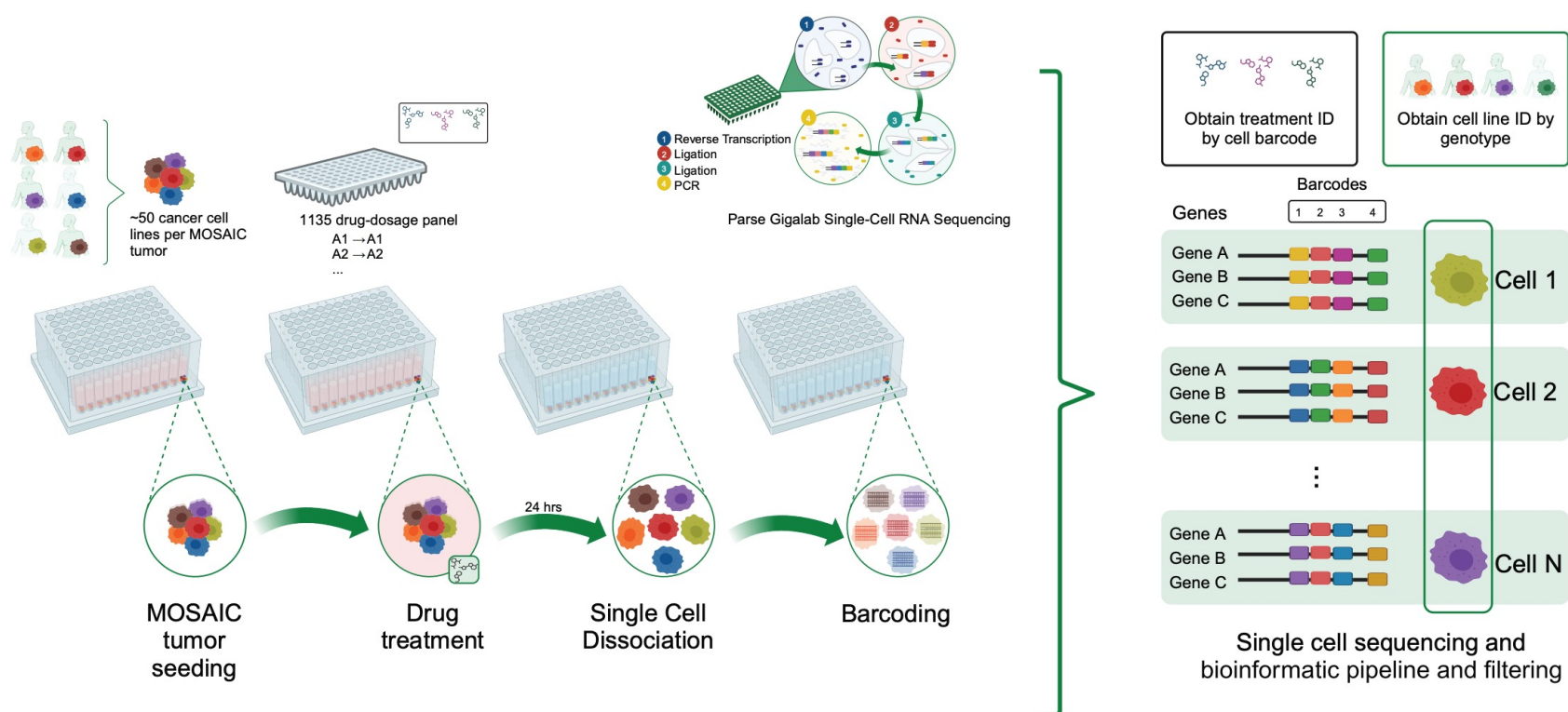
Three core goals:

- ① **Enable AI models** to learn context-dependent gene regulatory logic at scale
- ② **Map the perturbation landscape** across diverse genetic backgrounds and drug mechanisms
- ③ **Publicly release** the atlas to catalyze the systems biology and AI communities

The Mosaic Platform: Overview

A

Tahoe-100M Data Generation



Step-by-Step: From Cells to Atlas

- 1 **Cell Village Construction:** 50 cancer cell lines pooled into 3D spheroid "Mosaic tumors" per well; equal ratios using Cultrex scaffold in 14 × 96-well plates
- 2 **Drug Treatment:** 1,135 drug-dose combinations applied; 24 h incubation at 37°C, 5% CO₂; DMSO vehicle controls in wells H11/H12 on every plate
- 3 **Dissociation & Fixation:** TrypLE Express dissociation; Parse High-Throughput Evercode Cell Fixation v3; 96-well coordinate identity maintained throughout
- 4 **Combinatorial Barcoding:** Parse GigaLab kit — 3 rounds of split-pool ligation barcoding; ~100,000 cells per sublibrary; 1,786 total sublibraries
- 5 **Ultra-Scale Sequencing:** Ultima Genomics UG-100; ~1.4 trillion raw reads; 153 million cells pre-QC
- 6 **Bioinformatic Pipeline:** split-pipe → demuxlet SNP deconvolution → drug annotation → QC → AnnData

Atlas Scale at a Glance

100M+

Single-cell
transcriptomes
(95.6M passing full
QC)

50

Cancer cell lines
from 13 organs

1,135

Drug-dose
conditions
(379 unique drugs)

25

Mechanisms of
action
covered

1,287

Median cells per
cell line–drug–
dose

1.4T

Raw sequencing
reads
(Ultima UG-100)

31× more drug-perturbed conditions than SciPlex-3 | 29× more observations per condition than Replogle et al.

QC Strategy

Filter Level	Key Criteria
Full filters (primary analysis)	≥700 UMIs · <20% mito reads · UMI z-score ±3 · ≥250 genes · singlet only
Condition-level	≥50 cells per cell line–drug–dose condition; 3 low-abundance lines excluded

Biological Insights: Cellular Heterogeneity

Tahoe-100M reveals drug response heterogeneity at three levels:

- **Subpopulation dynamics:** Even within a single cell line, cells in different cell cycle states respond differently to the same drug — bulk assays would miss this entirely.
- **Pathway-specific effects:** BRAF inhibitors work only in BRAF-mutant cells; RAS inhibitors primarily in RAS-mutant cells. Mutation context is decisive.
- **Predictive biomarkers:** Transcriptional correlates of drug response across 50 lines can identify patient stratification signatures — linking cell line pharmacology to clinical prediction.

Understanding how treatment resistance emerges requires the full *distribution* of single-cell responses — not an average. Tahoe-100M provides exactly that.

Limitations and Future Directions

Current limitations

- **Cancer cell lines only** — in vitro models; tumor microenvironment absent
- **Pharmacological perturbations** — no genetic (CRISPR) perturbations in this release
- **24-hour endpoint** — early transcriptional response; chronic effects not captured
- **Fixed cells** — cannot capture real-time dynamics or live-cell imaging
- **SNP demultiplexing** — doublets at high cell density can be misclassified

Future directions

- Extend to **primary patient cells** and organoids
- Integrate **genetic perturbations** (Perturb-seq) with drug screens
- Add **multi-omic** modalities: ATAC-seq, proteomics
- Longer time-course measurements
- **AI model benchmarking** on Tahoe-100M as a community resource
- Combination drug treatment screens

Data available for download at **Arc Virtual Cell Atlas** (released February 25, 2025) · doi: 10.1101/2025.02.20.639398

Summary: Six Key Points

- 1 Scale** >100M single-cell transcriptomes — largest public perturbation scRNA-seq dataset
- 2 Platform** Mosaic cell villages + Parse GigaLab eliminates batch effects; Ultima Genomics delivers unprecedented throughput
- 3 Diversity** 50 cancer cell lines × 13 organs × 379 drugs × 25 MOAs × 3 doses = 52,886 unique conditions
- 4 Biology** Mutation-stratified drug responses (RAS/RAF), cell cycle arrest signatures, MOA-specific programs all validated
- 5 Quality** No detectable batch effects · Replicate Pearson $r \approx 0.975$ · >98% demultiplexing accuracy
- 6 Open resource** Publicly released to catalyze AI virtual cell modeling, precision oncology, and systems biology

THANK YOU